

Observational Foundations of Probability

A Representation Theorem for Observable Experiments

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Abstract

Many probabilistic theories begin by postulating an underlying probability space and then studying the observables derived from it. This work reverses that perspective. Starting from observable experiments organized as a refinement system of queries, we establish conditions under which compatible observable laws admit a canonical probability space representation.

We model observations as measurable queries with associated outcome spaces organized by a refinement preorder. From this system of queries we construct a canonical realization space as the projective limit of the observable outcome spaces. The observable σ -field is then generated by query-cylinder events.

Within this framework we establish two foundational results. First, probability measures on the observable σ -field are uniquely determined by the distributions of a generating family of queries, yielding an observational analogue of the classical π - λ uniqueness theorem. Second, compatible families of observable laws determine a canonical finitely additive observational content on cylinder events, and this content admits a probabilistic completion on the realization space by two independent routes.

The *realizability route* (Section 6) takes countably additive marginals as input and uses a realizability condition on Ω to produce the global measure via a direct Carathéodory argument, with no topology required on the outcome spaces. The *Prokhorov route* (Section 7) starts from only finitely additive marginals and derives countable additivity from a projective mass-concentration condition (surjective projective uniform tightness), at the cost of topological structure on the outcome spaces. Both routes yield a canonical probability space $(\Omega, \sigma(\mathcal{Q}), P)$ whose evaluation marginals recover the observable laws; under shared hypotheses they produce the same measure.

The resulting framework connects classical measure-theoretic probability, statistical experiment theory, and modern observable approaches to complex systems, and provides a foundation for extensions involving predictive structure, dynamical observables, and approximate compatibility.

1 Introduction and motivation

Probability theory traditionally begins with an underlying probability space and studies the observables derived from it. The present work reverses this perspective by starting from observable experiments themselves and asking when compatible observable laws admit a canonical probability space representation.

In many contemporary scientific settings the underlying event algebra is not directly accessible. Instead, systems are accessed through a family of admissible measurements, projections, or coarse-grainings. These observables determine the empirical distinctions that can be made about the system, while the latent event space remains implicit or even unknown. This raises a structural question: when can probabilistic structure be reconstructed from observable experiments alone?

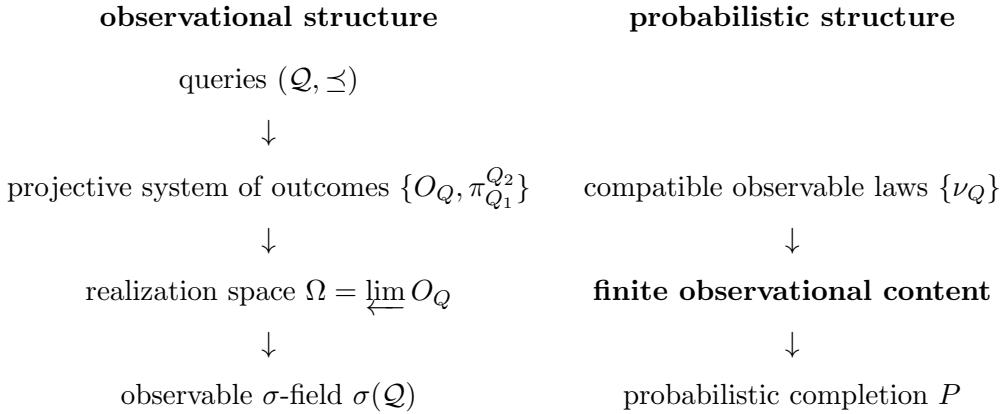
Problem. When do observable experiments determine an underlying probability space?

Answer. We show that compatibility of observable laws is precisely the condition required for such a representation, yielding a canonical probability space whose evaluation marginals recover the observable experiments. The paper establishes this by two complementary routes. The realizability route requires: (i) measurable structure on each query outcome space; (ii) compatible countably additive probability laws; (iii) directedness conditions; and (iv) realizability of the projective limit. The Prokhorov route replaces hypothesis (ii) by finitely additive laws satisfying surjective projective uniform tightness, and derives σ -additivity from topological compactness of the outcome spaces. These two routes resolve the extension problem from opposite directions: one assumes countable coherence, the other derives it.

Program perspective. This work is part of a broader program developing dynamical structure from observable experiments. The present paper establishes the first stage: the representation theorem.

Classically, Kolmogorov's extension theorem constructs a probability measure from consistent finite-dimensional marginals. The present construction plays an analogous role for observable experiments: the query system determines a projective family of observable laws whose compatibility yields the canonical experiment $(\Omega, \mathcal{F}, P)$.

The structure of the framework is summarized by the following three-layer architecture.



Main theorem. The principal result of this work establishes that compatible families of observable experiments admit a canonical probabilistic representation.

Theorem 1 (Observational representation theorem). *Let $(\mathcal{Q}, \preceq)$ be a query system equipped with a compatible family of observable laws (under the structural hypotheses of Sections 3 and 6: measurable outcome spaces, countably additive laws, directedness, and realizability). Then there exists a canonical probability space*

$$(\Omega, \mathcal{F}, P)$$

such that for every query $Q \in \mathcal{Q}$ the evaluation map

$$Q : \Omega \rightarrow O_Q$$

recovers the corresponding observable law.

Relation to the predictive sequel. The canonical experiment constructed here provides the probabilistic representation of observable experiments. A subsequent work studies how predictive state spaces and operator dynamics arise naturally from this observational representation.

Scope. The present paper does not derive local measurable structure or local countable additivity from observation alone. Rather, it shows that once queries are equipped with measurable outcome spaces and compatible countably additive observable laws, the remaining global probabilistic structure is canonically forced under explicit structural assumptions on the query system. The derivation of local probabilistic structure from more primitive observable distinctions is a direction for future work.

Structure of the paper. Section 2 recalls the classical uniqueness and extension principles that motivate the construction. Section 3 introduces query systems, their realization spaces, and the observable σ -field, together with the three structural conditions used in the main results. Section 4 proves the observational determination theorem. Section 5 constructs the finite observational content from compatible observable laws. Section 6 proves the observational extension theorem by a direct Carathéodory argument, establishing the canonical probability space $(\Omega, \sigma(\mathcal{Q}), P)$ without topological assumptions on the outcome spaces. Section 7 establishes the Prokhorov extension theorem: a parallel route in which σ -additivity is derived rather than assumed, under surjective projective uniform tightness of the marginals. The final sections interpret the framework in relation to statistical experiment theory and outline directions for predictive and dynamical extensions.

2 Classical uniqueness and extension

A central structural feature of classical probability theory is that a probability measure is determined by its values on a generating class of events. One of the most widely used tools expressing this fact is the π - λ theorem [?, ?]. Let $(\Omega, \mathcal{F})$ be a measurable space, and let $\mathcal{C} \subset \mathcal{F}$ be a π -system, that is, a collection of sets closed under finite intersections. If $\mathcal{C}$ generates $\mathcal{F}$, then any two probability measures P and P' satisfying

$$P(A) = P'(A), \quad A \in \mathcal{C},$$

necessarily agree on the full σ -algebra:

$$P = P' \text{ on } \mathcal{F}.$$

In this sense, the global probabilistic structure is determined by its values on a suitable generating class.

A particularly important application of this principle is Kolmogorov's extension theorem [?, ?]. In the classical construction of stochastic processes, one begins by specifying a single-time state space $(S, \mathcal{S})$ together with an index set T , typically representing time. The trajectory space is then taken to be the product

$$\Omega = S^T,$$

equipped with the product σ -algebra

$$\mathcal{F} = \mathcal{S}^{\otimes T}.$$

The measurable structure is generated by the coordinate projections

$$\pi_t : \Omega \rightarrow S, \quad \pi_t(\omega) = \omega(t).$$

More generally, for finite subsets $\{t_1, \dots, t_n\} \subset T$, one considers the joint projections

$$\pi_{t_1, \dots, t_n}(\omega) = (\omega(t_1), \dots, \omega(t_n)),$$

which generate the cylinder σ -field.

A stochastic process is then specified by a compatible family of finite-dimensional distributions

$$\mu_{t_1, \dots, t_n} \in \mathcal{P}(S^n),$$

satisfying the marginal consistency condition

$$(\pi_{t_1, \dots, t_n})_{\#} \mu_{t_1, \dots, t_m} = \mu_{t_1, \dots, t_n}, \quad n < m.$$

Kolmogorov's theorem asserts that such a compatible family determines a unique probability measure

$$P \in \mathcal{P}(\Omega, \mathcal{F})$$

whose finite-dimensional marginals coincide with the prescribed laws.

This construction makes a specific structural assumption about the organization of observational information. The starting point is a global trajectory space together with a canonical system of coordinate projections. Measurable events are generated by restrictions of these coordinates, and probabilistic structure is built relative to this coordinate representation.

However, in many contemporary scientific and statistical settings, this assumption is neither natural nor directly available. Systems are typically accessed through families of admissible measurements, projections, or coarse-grainings, without prior knowledge of a global coordinate structure. The observable interface is therefore given by measurable maps rather than by coordinates of an assumed latent space.

This observation suggests a reformulation in which admissible observations are taken as primitive. Instead of postulating a measurable event space in advance, we derive the observable σ -field from a family of measurable maps and formulate probabilistic structure in terms of compatibility of their induced laws.

3 Observational generators

Replacing coordinates by a more general notion of observable structure requires specifying the observational interface directly. Rather than beginning with a fixed measurable state space, we start from a family of admissible queries.

Definition 1 (Query). A *query* Q consists of a measurable outcome space

$$(O_Q, \mathcal{B}(O_Q)).$$

Remark 1 (Primitive structure). In the present paper the measurable structure $\mathcal{B}(O_Q)$ on each outcome space is taken as a primitive input. It represents the class of events distinguishable at resolution Q . A more foundational treatment would derive this σ -algebra from a designated class $\mathcal{E}_Q$ of admissible observable distinctions via $\mathcal{B}(O_Q) = \sigma(\mathcal{E}_Q)$; we regard this as a direction for future work.

A family of queries $\mathcal{Q}$ represents the admissible observational interface through which the system is accessed.

Definition 2 (Query system). A *query system* consists of:

1. a preorder $(\mathcal{Q}, \preceq)$;
2. for each $Q \in \mathcal{Q}$, a measurable outcome space

$$(O_Q, \mathcal{B}(O_Q));$$

3. for each relation $Q_1 \preceq Q_2$, a measurable refinement map

$$\pi_{Q_1}^{Q_2} : O_{Q_2} \rightarrow O_{Q_1},$$

satisfying the coherence relations

$$\pi_Q^Q = \text{id}_{O_Q}, \quad \pi_{Q_1}^{Q_2} \circ \pi_{Q_2}^{Q_3} = \pi_{Q_1}^{Q_3} \quad (Q_1 \preceq Q_2 \preceq Q_3).$$

In this way a query system $(\mathcal{Q}, \preceq)$ may be interpreted as an information order in the sense of Blackwell and Le Cam [?, ?]: queries higher in the order are at least as informative as those below them, since the outcomes of coarser queries are obtained from finer ones by measurable post-processing. Thus the refinement maps are coarsening maps: whenever $Q_1 \preceq Q_2$, the more informative query Q_2 determines the outcome of the less informative query Q_1 .

We will use two structural hypotheses, depending on the goal.

Definition 3 (Common coarsenings / lower-directedness). We say $(\mathcal{Q}, \preceq)$ admits *common coarsenings* if for any $Q_1, Q_2 \in \mathcal{Q}$ there exists $Q_3 \in \mathcal{Q}$ such that

$$Q_3 \preceq Q_1 \quad \text{and} \quad Q_3 \preceq Q_2.$$

Intuitively, Q_3 is a single (less informative) query whose outcomes can be pushed forward to recover the outcomes of both Q_1 and Q_2 . This is the assumption that any two observational resolutions admit a joint coarser representation.

Definition 4 (Common refinements / upper-directedness). We say $(\mathcal{Q}, \preceq)$ admits *common refinements* if for any $Q_1, Q_2 \in \mathcal{Q}$ there exists $Q_3 \in \mathcal{Q}$ such that

$$Q_1 \preceq Q_3 \quad \text{and} \quad Q_2 \preceq Q_3.$$

This is the assumption that any two observational resolutions can be jointly refined by a third — an assumption about the richness of the observational interface.

Definition 5 (Sequential common refinements). We say $(\mathcal{Q}, \preceq)$ admits *sequential common refinements* if every countable family $Q_1, Q_2, \dots \in \mathcal{Q}$ admits a common refinement: there exists $Q_* \in \mathcal{Q}$ such that

$$Q_n \preceq Q_* \quad \text{for all } n \geq 1.$$

This is the assumption that the observational interface is closed under countable joint refinement: however many query resolutions are specified, a single finer resolution exists that subsumes all of them. This is a substantive condition on the interface structure, not merely a technical regularity hypothesis.

Sequential common refinements imply common refinements (take the pair Q_1, Q_2 as the sequence). The sequential condition is strictly stronger and is the key analytic hypothesis enabling σ -subadditivity of the cylinder premeasure.

Definition 6 (Realizability). The query system is *realizable* if every local outcome extends to a global realization: for every $Q \in \mathcal{Q}$ and every $o \in O_Q$, there exists $\omega \in \Omega$ such that $\text{eval}_Q(\omega) = o$.

Equivalently, the evaluation maps $\text{eval}_Q : \Omega \rightarrow O_Q$ are all surjective.

Remark 2 (Status of realizability). Realizability is a substantive assumption. It asserts that the projective limit Ω is non-degenerate in a strong pointwise sense: every locally consistent outcome extends to a globally coherent realization. This is used in an essential way in the premeasure well-definedness argument and in the σ -subadditivity step of the extension theorem — it is not a side condition. Without it, both steps fail.

A natural measure-theoretic weakening — likely sufficient for the proof — is *almost-sure realizability*: for each Q and ν_Q -almost every $o \in O_Q$, there exists $\omega \in \Omega$ with $\text{eval}_Q(\omega) = o$. We retain the stronger pointwise form here for clarity. A sufficient condition for full realizability is given in Corollary 6.1: when each O_Q is Polish, refinement maps are continuous, and the index set is countable, the projective limit is Polish and realizability holds automatically.

Both structural hypotheses are used in the main results. Common coarsenings are used for the uniqueness theorem (the cylinder family forms a π -system). Common refinements are used for the extension theorem (finite cylinder families can be compressed to a single query level to define a well-posed premeasure). A sequential strengthening of common refinements is the key analytic condition that allows the premeasure to be extended to a full probability measure.

The space of realizations is defined as the set of coherent answers to all queries:

$$\Omega = \left\{ x \in \prod_{Q \in \mathcal{Q}} O_Q \mid x_{Q_1} = \pi_{Q_1}^{Q_2}(x_{Q_2}) \text{ whenever } Q_1 \preceq Q_2 \right\}.$$

Equivalently,

$$\Omega = \varprojlim_{Q \in \mathcal{Q}} O_Q,$$

the projective limit of the system of outcome spaces with bonding maps $\pi_{Q_1}^{Q_2}$ [?].

Remark 3 (Nondegeneracy of Ω). The realization space Ω is a derived object: it is determined entirely by the query system and carries no structure beyond what the refinement maps force. It is not assumed as a latent state space. However, Ω can be empty or degenerate without additional hypotheses on the query system — realizability (Definition 6) is precisely the condition asserting non-degeneracy.

The realization space Ω serves as a canonical space of coherent answers to all admissible queries. Probability measures on Ω represent probabilistic models compatible with the observable interface.

Each query Q induces an evaluation map

$$\text{eval}_Q : \Omega \rightarrow O_Q, \quad \text{eval}_Q(x) = x_Q.$$

Probability measures on Ω represent observational laws compatible with all admissible queries; the pushforwards $(\text{eval}_Q)_\# P$ recover the observable marginal distributions.

The distribution of a query under a probability measure $P \in \mathcal{P}(\Omega, \sigma(\mathcal{Q}))$ is given by the pushforward measure

$$Q_\# P = (\text{eval}_Q)_\# P \in \mathcal{P}(O_Q).$$

Lemma 1 (Projection compatibility). *For every relation $Q_1 \preceq Q_2$ in $\mathcal{Q}$ the evaluation maps satisfy*

$$\text{eval}_{Q_1} = \pi_{Q_1}^{Q_2} \circ \text{eval}_{Q_2}.$$

Proof. For $x \in \Omega$ we have

$$(\pi_{Q_1}^{Q_2} \circ \text{eval}_{Q_2})(x) = \pi_{Q_1}^{Q_2}(x_{Q_2}).$$

By the defining compatibility condition of Ω ,

$$x_{Q_1} = \pi_{Q_1}^{Q_2}(x_{Q_2}),$$

so the right-hand side equals $\text{eval}_{Q_1}(x)$. □

The observable σ -field is defined as the smallest σ -algebra making all evaluation maps measurable. Equivalently, it is generated by the cylinder sets

$$\text{Cyl}(Q, A) = \{x \in \Omega : \text{eval}_Q(x) \in A\}, \quad A \in \mathcal{B}(O_Q).$$

We denote this σ -field by

$$\sigma(\mathcal{Q}).$$

Lemma 2 (Measurability of evaluation maps). *For every query $Q \in \mathcal{Q}$, the evaluation map*

$$\text{eval}_Q : \Omega \rightarrow O_Q$$

is measurable with respect to $(\Omega, \sigma(\mathcal{Q}))$.

Proof. By definition $\sigma(\mathcal{Q})$ is the smallest σ -algebra making all evaluation maps measurable. $\square$

The common-coarsening hypothesis introduced above is a meet-like property of the information order on queries. Intuitively, it ensures that finite collections of observable constraints can be represented at a single query level.

Lemma 3 (Cylinder π -system). *Assume $(\mathcal{Q}, \preceq)$ admits common coarsenings in the sense of Definition 3.*

Then the family of cylinder sets

$$\mathcal{C}_{\mathcal{Q}} = \{\text{Cyl}(Q, A) : Q \in \mathcal{Q}, A \in \mathcal{B}(O_Q)\}$$

forms a π -system.

Proof. Let $\text{Cyl}(Q_1, A_1)$ and $\text{Cyl}(Q_2, A_2)$ be two cylinders. The common-coarsening hypothesis provides a query Q_3 such that

$$Q_3 \preceq Q_1 \quad \text{and} \quad Q_3 \preceq Q_2.$$

Using the refinement maps $\pi_{Q_3}^{Q_1}$ and $\pi_{Q_3}^{Q_2}$, a direct calculation shows

$$\text{Cyl}(Q_1, A_1) \cap \text{Cyl}(Q_2, A_2) = \text{Cyl}\left(Q_3, (\pi_{Q_3}^{Q_1})^{-1}(A_1) \cap (\pi_{Q_3}^{Q_2})^{-1}(A_2)\right).$$

Since measurable sets are stable under preimages and intersections, the right-hand side is again a cylinder. $\square$

Remark 4 (Relation to classical coordinate constructions). In the classical Kolmogorov construction of stochastic processes, the indexing family of coordinate projections is ordered by inclusion of finite coordinate sets and is therefore upper-directed. The observational framework uses the opposite side of the information order: common coarsenings permit finite collections of observable constraints to be represented at a single query level.

4 Observational determination

The following result is the observational analogue of the classical π - λ uniqueness theorem.

Theorem 2. *Let $(\mathcal{Q}, \preceq)$ admit common coarsenings (Definition 3) and let P and P' be probability measures on the observable space $(\Omega, \sigma(\mathcal{Q}))$ generated by $\mathcal{Q}$.*

If

$$Q_{\#}P = Q_{\#}P'$$

for all $Q \in \mathcal{Q}$, then $P = P'$.

Proof. Equality of pushforward laws implies that P and P' agree on all cylinder events. By Lemma 3 the cylinder family forms a π -system generating $\sigma(\mathcal{Q})$. The classical π - λ theorem therefore implies that the two measures coincide on the entire observable σ -field. $\square$

Remark 5. The determination theorem is purely measure-theoretic. It depends only on the observable σ -field generated by queries and on the lower-directedness hypothesis ensuring that cylinder events form a π -system. No compactness or topological assumptions enter at this stage.

5 Finite observational content

Definition 7 (Cylinder set semiring). Let

$$\mathcal{C}_{\mathcal{Q}} = \{\text{Cyl}(Q, A) : Q \in \mathcal{Q}, A \in \mathcal{B}(O_Q)\}$$

denote the family of cylinder sets.

Lemma 4 (Cylinder set semiring structure). *Assume $(\mathcal{Q}, \preceq)$ admits common coarsenings (Definition 3). Then $\mathcal{C}_{\mathcal{Q}}$ is a set semiring: it is closed under finite intersections (already the π -system property of Lemma 3) and set differences decompose as finite disjoint unions of cylinders.*

Proof sketch. Intersection: Lemma 3. Difference: given $\text{Cyl}(Q_1, A) \setminus \text{Cyl}(Q_2, B)$, use a common coarsening $Q_3 \preceq Q_1, Q_2$ to write both as cylinders at Q_3 ; the difference is then a single cylinder at Q_3 and is itself in $\mathcal{C}_{\mathcal{Q}}$. $\square$

Lemma 5 (Finite cylinder compression). *Assume $(\mathcal{Q}, \preceq)$ admits common refinements (Definition 4).*

Then for any finite family of queries $Q_1, \dots, Q_n$ and measurable sets $A_i \in \mathcal{B}(O_{Q_i})$, there exists a common refinement Q_ such that*

$$\bigcap_{i=1}^n \text{Cyl}(Q_i, A_i) = \text{Cyl}\left(Q_*, \bigcap_{i=1}^n (\pi_{Q_i}^{Q_*})^{-1}(A_i)\right).$$

Proof. Upper-directedness gives a common refinement Q_* with $Q_i \preceq Q_*$ for all i . Then $\pi_{Q_i}^{Q_*} : O_{Q_*} \rightarrow O_{Q_i}$ is well-typed and $\text{Cyl}(Q_i, A_i) = \text{Cyl}(Q_*, (\pi_{Q_i}^{Q_*})^{-1}(A_i))$ by the refinement identity. The intersection follows. $\square$

Remark 6. Lemma 5 uses common *refinements* rather than common coarsenings. The direction matters: the projection $\pi_{Q_i}^{Q_*} : O_{Q_*} \rightarrow O_{Q_i}$ goes from the finer space to the coarser one, so it is the preimage that lives in O_{Q_*} and the constraint set is well-typed. An upper bound in the information order is what is needed here.

We now turn to the question of probabilistic realization. Suppose we are given, for each query $Q \in \mathcal{Q}$, a probability law

$$\nu_Q \in \mathcal{P}(O_Q).$$

Definition 8 (Observational compatibility). A family $\{\nu_Q\}_{Q \in \mathcal{Q}}$ with $\nu_Q \in \mathcal{P}(O_Q)$ is *compatible* if for every relation $Q_1 \preceq Q_2$ in the query preorder,

$$(\pi_{Q_1}^{Q_2})_{\#} \nu_{Q_2} = \nu_{Q_1}.$$

In other words, whenever one query is obtained from another by coarsening, the corresponding probability laws agree under the induced marginalization.

Theorem 3 (Finite observational content). *Suppose:*

1. *The refinement preorder on $\mathcal{Q}$ admits common coarsenings (Definition 3) and common refinements (Definition 4).*
2. *The query system is realizable (Definition 6).*
3. *The family $\{\nu_Q\}_{Q \in \mathcal{Q}}$ is observationally compatible in the sense of Definition 8.*

Then the family $\{\nu_Q\}$ determines a well-defined finitely additive content μ_0 on $\mathcal{C}_{\mathcal{Q}}$:

$$\mu_0(\text{Cyl}(Q, A)) := \nu_Q(A), \quad A \in \mathcal{B}(O_Q).$$

Proof sketch. Well-definedness. Suppose $\text{Cyl}(Q, A) = \text{Cyl}(Q', A')$. Use common refinements to find Q_* finer than both Q and Q' . The constraint sets at Q_* are the preimages of A and A' under $\pi_Q^{Q_*}$ and $\pi_{Q'}^{Q_*}$ respectively. Since $\text{Cyl}(Q, A) = \text{Cyl}(Q', A')$, surjectivity of eval_{Q_*} (realizability) implies the two preimage sets in O_{Q_*} coincide. Compatibility then gives $\nu_Q(A) = \nu_{Q_*}(\text{preimage}) = \nu_{Q'}(A')$.

Finite additivity. For disjoint cylinders with measurable union, the same compression to a common refinement level Q_* reduces additivity to that of ν_{Q_*} . $\square$

Remark 7. The passage from finite content to a countably additive probability measure on $(\Omega, \sigma(\mathcal{Q}))$ requires one further condition: σ -subadditivity of μ_0 . This is the analytic heart of the extension problem, and the next section shows how sequential common refinements supply it.

Hypothesis (3) requires each ν_Q to be a countably additive probability measure. This is assumed, not derived: the countable additivity present in the global extension theorem (§6) is propagated from this local assumption via the compression argument, not constructed from below. The question of whether countable additivity of the ν_Q can itself be derived from compatibility and interface structure alone is a direction for future work.

6 Observational extension theorem

The finite content μ_0 constructed in the previous section is finitely additive on the cylinder set semiring $\mathcal{C}_{\mathcal{Q}}$. The central result of this paper shows that sequential common refinements supply the missing σ -subadditivity, allowing Carathéodory's extension theorem [?, ?] to produce a full probability measure on $(\Omega, \sigma(\mathcal{Q}))$.

Theorem 4 (Observational extension theorem). *Suppose:*

1. $(\mathcal{Q}, \preceq)$ admits common coarsenings (Definition 3).
2. $(\mathcal{Q}, \preceq)$ admits sequential common refinements (Definition 5).
3. The query system is realizable (Definition 6).
4. The family $\{\nu_Q\}_{Q \in \mathcal{Q}}$ is observationally compatible (Definition 8) and each ν_Q is a probability measure.

Then there exists a unique probability measure

$$P \in \mathcal{P}(\Omega, \sigma(\mathcal{Q}))$$

such that for every $Q \in \mathcal{Q}$,

$$(\text{eval}_Q)_\# P = \nu_Q.$$

Proof. Step 1: Finite content. Theorem 3 gives a well-defined finitely additive content μ_0 on $\mathcal{C}_{\mathcal{Q}}$ with $\mu_0(\text{Cyl}(Q, A)) = \nu_Q(A)$. By Lemma 4, $\mathcal{C}_{\mathcal{Q}}$ is a set semiring.

Step 2: σ -subadditivity. Let $\text{Cyl}(Q, B) \subseteq \bigcup_{n \geq 1} \text{Cyl}(Q_n, A_n)$ with all sets in $\mathcal{C}_{\mathcal{Q}}$. Apply sequential common refinements to the sequence $Q, Q_1, Q_2, \dots$ to obtain a single query Q_* with $Q \preceq Q_*$ and $Q_n \preceq Q_*$ for all n .

Compression (Lemma 5) rewrites each cylinder at level Q_* :

$$\text{Cyl}(Q, B) = \text{Cyl}(Q_*, E), \quad \text{Cyl}(Q_n, A_n) = \text{Cyl}(Q_*, E_n),$$

where $E = (\pi_Q^{Q_*})^{-1}(B)$ and $E_n = (\pi_{Q_n}^{Q_*})^{-1}(A_n)$ are measurable subsets of O_{Q_*} . Realizability and the set inclusion in Ω imply $E \subseteq \bigcup_n E_n$ in O_{Q_*} . Therefore

$$\mu_0(\text{Cyl}(Q, B)) = \nu_{Q_*}(E) \leq \sum_{n=1}^{\infty} \nu_{Q_*}(E_n) = \sum_{n=1}^{\infty} \mu_0(\text{Cyl}(Q_n, A_n)),$$

where the inequality is σ -subadditivity of the single measure ν_{Q*} .

Step 3: Carathéodory extension. Since $\mathcal{C}_Q$ is a set semiring generating $\sigma(\mathcal{Q})$ and μ_0 is a σ -subadditive additive content on it, Carathéodory's extension theorem [?, ?] applies and yields a measure P on $(\Omega, \sigma(\mathcal{Q}))$ extending μ_0 .

Step 4: Marginal recovery. For any $A \in \mathcal{B}(O_Q)$,

$$P(\text{eval}_Q^{-1}(A)) = P(\text{Cyl}(Q, A)) = \mu_0(\text{Cyl}(Q, A)) = \nu_Q(A),$$

so $(\text{eval}_Q)_\# P = \nu_Q$.

Step 5: P is a probability measure. $P(\Omega) = P(\text{Cyl}(Q, O_Q)) = \nu_Q(O_Q) = 1$ for any Q .

Uniqueness. Two probability measures with the same evaluation marginals agree on every cylinder event, hence on $\sigma(\mathcal{Q})$ by Theorem 2. $\square$

Remark 8 (Assumptions versus derived structure). The theorem assumes: (i) measurable structure on each outcome space; (ii) countable additivity of each observable law ν_Q ; (iii) lower-directedness, sequential upper-directedness, and realizability of the query system. It derives: a canonical σ -additive probability measure on $(\Omega, \sigma(\mathcal{Q}))$ whose evaluation marginals recover all observable laws.

Countable additivity of P is not constructed from first principles but propagated from the countable additivity of ν_{Q*} via the compression argument. Assumptions (i) and (ii) are inputs from classical measure theory; a more foundational treatment would aim to derive them from the structure of observable distinctions. Assumptions (iii) are substantive conditions on the query system: they express joint coarsenability, closure under countable joint refinement, and non-degeneracy of the realization space.

Remark 9 (Structure of the proof). The proof uses the two directedness conditions for distinct roles. Common coarsenings give the semiring structure on $\mathcal{C}_Q$ (needed for Carathéodory) and the π -system structure needed for uniqueness. Sequential common refinements compress any countable cylinder family to a single query level, reducing σ -subadditivity of μ_0 to σ -subadditivity of the single measure ν_{Q*} , which is immediate. Realizability is used only to translate set inclusions in Ω to set inclusions in O_{Q*} .

Remark 10 (Relation to Kolmogorov's extension theorem). Kolmogorov's theorem constructs a probability measure on S^T from consistent finite-dimensional distributions. The index set is a coordinate product and the consistency condition is the push-forward condition under coordinate projections.

The observational extension theorem plays the same role for query systems. The realization space $\Omega = \varprojlim O_Q$ replaces the product S^T , compatible marginals replace consistent finite-dimensional distributions, and sequential common refinements replace the countable-cofinality condition that reduces the Kolmogorov argument to a countable system.

The key conceptual difference is that the observational route imposes no topology on the outcome spaces. The entire construction is measure-theoretic: the measure P is built directly on the $\sigma(\mathcal{Q})$ -measurable structure of Ω by Carathéodory's theorem, without any appeal to tightness or compactness.

Corollary 1 (Topological setting). *In particular, Corollary 1 provides the main route to verifying realizability: under Polish outcome spaces and a countable cofinal chain, all four hypotheses of Theorem 4 are automatically satisfied.*

The hypotheses of Theorem 4 are satisfied whenever:

1. *each outcome space O_Q is Polish and the refinement maps are continuous (so Ω is a closed subspace of a countable product of Polish spaces, hence Polish [?] and realizable);*

2. the query preorder contains a countable cofinal chain $Q_1 \preceq Q_2 \preceq \dots$ (so every sequence of queries admits a common refinement within this chain).

In this case the compatible observable laws need not be assumed Radon: the measure P produced by Theorem 4 is automatically a Radon probability measure on the Polish space Ω .

Remark 11 (Lean formalization). The proof of Theorem 4 given above is the proof that is fully formalized in Lean 4 [?] (`observational_extension` in `QuerySystem.lean`, 0 sorrys). The Lean hypotheses correspond exactly to the four conditions of the theorem: `LowerDirected` (common coarsenings), `SequentiallyUpperDirected` (sequential common refinements), `EvalSurjective` (realizability), and `CompatibleMarginals` with `IsProbabilityMeasure`. The Carathéodory step uses `AddContent.measure` from Mathlib [?].

7 Prokhorov extension for query systems

Theorem 4 proceeds from the *realizability* side: given σ -additive marginals $\{\nu_Q\}$ and a realizability condition on Ω , it produces the global measure P . This section establishes a route from the opposite direction, the *tightness* side: given only *finitely* additive marginals, a projective mass-concentration condition forces σ -additivity and delivers P directly. These two theorems resolve the same extension problem from opposite directions: one assumes countable coherence, the other derives it.

Projective uniform tightness

Because the marginals $\{\nu_Q\}$ live on different spaces O_Q , the classical Prokhorov notion of tightness cannot be applied level by level. The correct formulation is projective: compact approximants must be chosen compatibly across levels so that they assemble into a compact subset of the inverse limit Ω .

Definition 9 (Surjective projective uniform tightness). A finitely compatible family $\{\nu_Q\}_{Q \in \mathcal{Q}}$ of finitely additive normalized measures on $(\mathcal{B}(O_Q))_{Q \in \mathcal{Q}}$ is *surjectively projectively uniformly tight* if for every $\varepsilon > 0$ there exists a system $\{K_Q\}_{Q \in \mathcal{Q}}$ satisfying:

1. $K_Q \subset O_Q$ is compact for each $Q \in \mathcal{Q}$,
2. $\pi_{Q_1}^{Q_2}(K_{Q_2}) = K_{Q_1}$ for all $Q_1 \preceq Q_2$ (*surjectivity of bonding maps on compact sets*),
3. $\nu_Q(O_Q \setminus K_Q) < \varepsilon$ for every $Q \in \mathcal{Q}$.

Condition (2) ensures the compact sets form a consistent inverse system, so their intersection in Ω ,

$$K_\Omega := \Omega \cap \prod_{Q \in \mathcal{Q}} K_Q = \varprojlim K_Q,$$

is nonempty and compact, and each projection $\text{eval}_Q : K_\Omega \rightarrow K_Q$ is surjective (by the inverse-limit theorem for compact Hausdorff spaces with surjective bonding maps).

Remark 12 (Relation to individual tightness). Definition 9 strengthens individual tightness (compact K_Q at each level independently) by requiring the compact approximants to be chosen compatibly under refinement with surjective bonding maps. This cross-level compatibility is precisely what is needed to lift a local nonemptiness argument into Ω .

The theorem

Theorem 5 (Prokhorov extension for query systems). *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system in which each outcome space O_Q is Hausdorff with $\mathcal{E}_Q = \mathcal{B}(O_Q)$, and each refinement map $\pi_{Q_1}^{Q_2} : O_{Q_2} \rightarrow O_{Q_1}$ is continuous. Let $\{\nu_Q\}_{Q \in \mathcal{Q}}$ be a finitely compatible family of finitely additive normalized measures on $\mathcal{B}(O_Q)$.*

If $\{\nu_Q\}$ is surjectively projectively uniformly tight (Definition 9), then there exists a unique σ -additive probability measure P on $(\Omega, \sigma(\mathcal{Q}))$ satisfying

$$(\text{eval}_Q)_\# P = \nu_Q \quad \text{for every } Q \in \mathcal{Q}.$$

Uniqueness is among all σ -additive probability measures on $(\Omega, \sigma(\mathcal{Q}))$ with the prescribed marginals.

Proof sketch. Step 1: Compact core. For each $\varepsilon > 0$ apply Definition 9 to obtain $\{K_Q\}$ with surjective bonding maps, compact $K_\Omega = \varprojlim K_Q$, and $\nu_Q(O_Q \setminus K_Q) < \varepsilon$ for all Q .

Step 2: Closed-cylinder continuity from above. Let $C_N = \text{Cyl}(Q_N, F_N) \downarrow \emptyset$ with each $F_N \subseteq O_{Q_N}$ closed. Assuming $\mu_0(C_N) \geq \varepsilon$ for all N (seeking contradiction), the mass estimate $\nu_{Q_N}(F_N \cap K_{Q_N}) \geq \varepsilon/2 > 0$ combined with surjectivity of $\text{eval}_{Q_N} : K_\Omega \rightarrow K_{Q_N}$ gives $C_N \cap K_\Omega \neq \emptyset$ for all N . Since F_N is closed and eval_{Q_N} is continuous, each $C_N \cap K_\Omega$ is closed in the compact space K_Ω . The finite intersection property yields $\bigcap_N (C_N \cap K_\Omega) \neq \emptyset$, contradicting $C_N \downarrow \emptyset$ in Ω .

Step 3: Extension to Borel cylinders. Each K_Q is compact Hausdorff, so $\nu_Q|_{K_Q}$ is regular by Alexandrov's theorem (no additional hypothesis required). Inner regularity approximates every Borel cylinder by closed cylinders to within $O(\varepsilon)$. Sequential upper-directedness provides common refinements for finite intersections of closed cylinders, preserving closedness. A standard approximation argument reduces the general Borel case to the closed-cylinder case already handled. Carathéodory's theorem then extends the σ -additive premeasure μ_0 uniquely to $\sigma(\mathcal{C}_\mathcal{Q}) = \sigma(\mathcal{Q})$ on the ambient product.

Step 4: Concentration on Ω (graph-support lemma). For each comparable pair $Q_1 \preceq Q_2$, the intersection $\text{Cyl}(Q_1, A) \cap \text{Cyl}(Q_2, B) = \text{Cyl}(Q_2, (\pi_{Q_1}^{Q_2})^{-1}(A) \cap B)$ shows the joint law of $(\text{eval}_{Q_1}, \text{eval}_{Q_2})$ under P equals $(\text{graph}_\pi)_\# \nu_{Q_2}$, which is supported on the graph $G_{Q_1, Q_2} = \{(a, b) : a = \pi_{Q_1}^{Q_2}(b)\}$. Hence the defect set $D_{Q_1, Q_2} := \{x : x_{Q_1} \neq \pi_{Q_1}^{Q_2}(x_{Q_2})\}$ has $P(D_{Q_1, Q_2}) = 0$. Since $\mathcal{Q}$ is countable there are countably many comparable pairs; a countable union of null sets gives $P(\Omega) = 1$. Restricting P to Ω yields the required measure. Uniqueness follows because cylinder sets generate $\sigma(\mathcal{Q})$ and determine any σ -additive probability measure with the given marginals. $\square$

Proposition 1 (Consistency between routes). *If $\{\nu_Q\}$ is a family of σ -additive probability measures satisfying both the hypotheses of Theorem 4 and those of Theorem 5, then both constructions produce the same measure P .*

Proof. Both measures are σ -additive on $(\Omega, \sigma(\mathcal{Q}))$ with marginals ν_Q for every Q . Since cylinder sets generate $\sigma(\mathcal{Q})$ and form a π -system, uniqueness of σ -additive extensions implies equality. $\square$

Remark 13 (Comparison of the two routes). The two extension theorems address complementary scenarios.

Route	Input	Key mechanism
Realizability (Thm. 4)	σ -additive ν_Q	realizability of Ω
Prokhorov (Thm. 5)	finitely additive ν_Q	projective tightness

The realizability route assumes countable coherence of the marginals and requires no topology on the outcome spaces. The Prokhorov route derives countable coherence from a topological mass-concentration condition: surjective projective compact cores prevent mass from escaping the inverse limit. Both routes conclude that a unique global measure P on $(\Omega, \sigma(\mathcal{Q}))$ exists with the prescribed marginals.

In practice, the realizability route applies when countable additivity of the marginals is known a priori; the Prokhorov route applies when only finite compatibility is available but surjective projective uniform tightness (Definition 9) can be verified.

The role of projective tightness in the absence of countability, and its relation to structural conditions on the query system, is discussed in Section 10.

8 Representation and experiment-theoretic interpretation

The preceding results show that the probabilistic content of a model is governed by observable regularities and their compatibility relations. Compatible observable marginals determine a canonical content on cylinder events; sequential common refinements allow this content to be extended to a full probability measure on the observational horizon. Different realizations may therefore support the same observational laws, which motivates the following notion.

The structural relation to the theory of statistical experiments may also be made explicit. In the sense of Blackwell and Le Cam [?, ?], each query together with its induced law defines an observable experiment, and the refinement preorder records the corresponding informativeness order under measurable post-processing. From this perspective, the present framework and classical experiment theory are built from closely related ingredients:

Le Cam experiment theory observational framework

statistical experiment	query
parameter space	realization space Ω
observation map	evaluation map
post-processing / garbling	query refinement
experiment family	query system
experiment comparison	informativeness preorder
experiment equivalence	observational equivalence

The difference lies in the direction of construction. Classical experiment theory begins with an underlying probabilistic model and asks how different observational mechanisms compare. The present framework begins with a compatible family of observable experiments and asks when they determine a probabilistic model. In this sense, Le Cam theory moves from models to experiments, whereas the present framework moves from experiments to models.

More concretely, in classical experiment theory a statistical experiment is given by a family

$$\theta \mapsto P_\theta.$$

In the present setting an observable experiment is given by a query

$$\omega \mapsto Q(\omega).$$

The refinement relation

$$Q_1 \preceq Q_2 \iff Q_1 = \pi \circ Q_2$$

is therefore the same structural relation that appears in the Blackwell–Le Cam theory of experiment comparison [?, ?, ?]: one experiment is obtained from another by measurable post-processing. The query system may thus be interpreted as a preorder of observable experiments ordered by informativeness.

Definition 10 (Observational equivalence). Two realizations are observationally equivalent if they induce the same pushforward laws for every admissible query.

Different measurable realizations may support the same observable laws. In particular, Ω is not a hidden state space postulated independently of the observable interface: it is the canonical consistency object determined by the query system, carrying exactly the structure forced by the refinement maps and no more. In that sense, the latent measurable space functions as a representation of the observable structure rather than as a primitive object. From this viewpoint the entire framework may be read as an inverse problem for experiment theory: instead of comparing experiments relative to a fixed model, one reconstructs the model from a compatible family of experiments.

This viewpoint is especially relevant for subsequent developments. There the basic progression is not merely

queries $\rightarrow$ predictive experiments $\rightarrow$ predictive sufficiency $\rightarrow$ observable operators,

but more specifically a passage from observable experiment structure to predictive experiment structure. The broader program may therefore be summarized as

experiment lattice $\rightarrow$ observable compatibility $\rightarrow$ probability,

so that probability appears as a representation theorem for observable information structures.

The predictive structure that emerges from the canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$ — predictive kernels, minimal predictive state spaces, and operator representations — is studied in subsequent work.

9 Approximate compatibility

In practical settings, exact compatibility is rarely satisfied. Instead, observational laws are typically consistent only up to controlled discrepancies. A key question is whether control of divergences

$$D(Q_{\#}P \parallel Q_{\#}P')$$

on generators implies global control. This suggests several investigative directions into robustness, optimal transport, and information geometry:

- Stability and approximate extension under bounds on divergences such as total variation, f -divergences, or Wasserstein metrics [?].
- Information-geometric formulations of observational distinguishability and compatibility.
- Robust probabilistic modeling in high-dimensional and partially observed systems.

These problems are closely related to contemporary themes in statistics and machine learning, including distributional robustness, uncertainty quantification, and inference under model misspecification. The observational perspective suggests that probabilistic structure may be viewed as the most stable completion of empirical regularities rather than as an exact underlying law.

10 Outlook and connections

The observational formulation developed here provides a unified viewpoint linking several classical and modern strands of probability and statistics:

1. Classical stochastic processes arise as a special case in which the observational interface consists of coordinate projections of a product space S^T . The observational extension theorem recovers Kolmogorov’s construction in this setting (via Corollary 1) while clarifying its structural content: the key hypothesis is a countable cofinality condition on the index set, not topology on the factor spaces.
2. Symmetry and exchangeability constraints on observational laws lead naturally to representation theorems such as de Finetti’s [?]. In this setting, mixture representations appear as structural consequences of invariance of the observational interface.
3. The theory of statistical experiments and sufficiency can be interpreted in terms of refinement and equivalence of observational systems. Minimal sufficient representations correspond to minimal observational structures preserving predictive content.
4. The framework connects naturally to dynamical and operator-theoretic approaches. In time-evolving systems, observables generate predictive structure, and Markov or semi-group properties may be interpreted as emergent regularities of compatible observational laws. This perspective suggests new connections to transfer operators, Koopman theory [?, ?], and data-driven dynamical modeling.
5. The observational viewpoint provides a potential synthesis of classical objectivist and Bayesian interpretations. Stable empirical regularities and coherent inference both arise as properties of compatible observational systems, without requiring a prior commitment to either ontological or subjective foundations.

These connections motivate further development of approximate, dynamical, and geometric extensions of the theory. In particular, the emergence of predictive structure, robustness under partial observability, and operator-based representations of observational consistency form key directions for future work.

Open frontier: structural forcing of σ -additivity. Section 7 derives σ -additivity from projective tightness in the countable setting. For uncountable query systems without a countable determining subsystem, concentration on Ω becomes a genuine obstruction: an uncountable intersection of full-measure defect-free sets need not have full measure.

The deeper open problem is whether σ -additivity can be forced from purely structural conditions on the query system $(\mathcal{Q}, \preceq, \{\pi_{Q_1}^{Q_2}\})$, without any assumption on the individual marginals. The Barrier theorem (that the refinement maps are inert with respect to continuity from above, since preimage preserves decreasing intersections exactly) rules out map-level non-degeneracy conditions. What remains is the question:

On query systems of what type is every finitely compatible family automatically σ -additive?

Current evidence points toward structural conditions that prevent persistent decreasing compatible tails from accumulating mass — the query-system analogue of Prokhorov’s tightness, but derived from system architecture rather than imposed on the marginals.

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